



Zoology Legends

Bio201

Mid_Term Papers Spring 2023

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Zoology Legends

Repeated MCQS

- The inner membrane of mitochondria is called the...**cristae**.
- Amino acids can be free (**unbound**) or attached to tRNA molecules during protein synthesis.
- The movement of electrons itself is represented by a...**negative sign**.
- **Endocytosis** is the process by which cells internalize substances from the external environment by engulfing them.
- Cells discovered by... **Antonie van Leeuwenhoek**
- Feature of quantitative PCR:

Answer: **Probe**

- Lipid molecules are held by intermolecular forces:

Answer: **Van der Waals forces**

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- Some bacteria have flagella made of:

Answer: **Protein**

- Cementation of cell is "cell adhesion":

Answer: **True**

- Cellular membrane acts as a barrier:

Answer: **True**

- Separation of centrosomes and movement towards the opposite pole:

Answer: Late **prophase**

- Energy due to motion:

Answer: **Kinetic energy**

- DNA to RNA, RNA to protein process is:

Answer: **Gene expression**

- Enzyme binding is related to:

Answer: **Substrate**

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- The tendency of an atom to attract electrons towards itself:

Answer: **Electronegativity**

- Flavin is:

Answer: **A Coenzyme**

- Translation takes place in the:

Answer: **Cytoplasm**

- A nucleotide is a simple sugar molecule bonded to a:

Answer: **Phosphate group**

- Denaturation occurs due to:

Answer: **Both pH and heat**

- In the **mitotic phase**, the cell undergoes mitosis, where the replicated chromosomes are separated into two identical daughter cells

- Not a phase of interphase:

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Answer: **Meiosis**

- Example of variable expressivity:

Answer: **Huntington**

- Simian crease is:

Answer: **Single palmar fold**

- Function of Golgi Apparatus:

Answer: **Protein modification and sorting**

- In mitosis, DNA molecule divides by:

Answer: **Equally**

- Dynein & Kinesin are classes of

Motor Proteins

- Enzyme that works at neutral conditions:

Answer: **Amylase**

- Fatty Acid Length:

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14-24 Cs

- Mendel's Second law was about:

Answer: **Independent assortment of alleles**

- The length of a primer in PCR typically ranges from **18 to 30 bases**.
- In their experiment, Beadle and Tatum added the amino acids **arginine, leucine, and methionine** to their culture.
- The spindles that are not attached to chromosomes are known as **astral microtubules**
- The three stop codons in the genetic code are **UAA, UAG, and UGA**.
- Macromolecules of cell are **4**
- hydrogen bonds are classified as **strong** intermolecular forces

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- In which phase does even distribution of centrosome occur and centrosomes move to opposite poles?...**Mitotic phase**
- Ribosomes are attached to **ER**
- **Copper** is a co Factor
- The classification of proteins as either globular or fibrous is based on their **structural characteristics**.
- Lipids are **insoluble** in water.
- The person who discovered the PCR (Polymerase Chain Reaction) technique is **Kary Mullis**.
- The molecule that converts light energy into chemical energy during photosynthesis is **chlorophyll**.
- Measuring unit of genetic makeup is **DNA**.
- **Cell division** is the basic process that ensures the continuity of life.
- An average globular cell has ----- mitochondria. **10,000**
- **Meiosis**: Organisms shuffle genes.

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- Untreated PKU can cause problems with brain development, leading to progressive **mental retardation** and seizures
- **T (transfer) site:** first site where charged tRNA lands with the help of transfer factor
- Mendel donated dominant and recessive traits as...**SS and ss**
- The scientist who proposed that DNA is the hereditary material is **Oswald Avery**.
- The appropriate primer length in PCR (Polymerase Chain Reaction) is usually around **18 to 22** nucleotides long.
- DNA replicates during...**S phase**
- At which Temperature denaturation occurs....**95**
- how many phenotypes was studied by Mendel?... **7**
- The hydrogen bond is a**Weak bond**
- Activation energy of an enzymatic reaction is... **Low**

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- The folding of a **protein polypeptide** chain takes place in the **tertiary structure**.
- **SEQUENCING**: Sequencing determines the base pair sequence of a gene.
- SER (Smooth Endoplasmic Reticulum) is **not involved** in protein synthesis. Its main function is the **synthesis of lipids**, steroids, and carbohydrates, as well as the metabolism of exogenous substances, such as drugs or toxins.
- HIV is a...**RNA virus**
- Endonucleases is another name for an enzyme...**restriction endonuclease**
- Tail is **hydrophobic**
- **Lipids** are **insoluble** in water.
- The role of non-polar amino acids outside the macromolecule is to provide **hydrophobic interactions**.

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- Aik kuch asy tha when nucleotide are arranged.. as result what we found equal fctor or non equal ?
....**Equal**
- What chromosome is larger X or Y?... **X**
- cell membrane expels partials by vesicle formation...**Exocytosis**
- Phenylalanine hydroxylase metabolites....**Phenylalanine to tyrosine**
- Which of the following stabilises cell shape...**Actin**
- The RNA is ... while DNA is ...**Single stranded and double stranded**
- Mendel's second law is ...**Independent assortment**
- Stitching two different DNAs together is ...**Genetic engineering**
- In a chemical reaction equilibrium is... **Static State**,
No Net Charge, and $\Delta G = 0$
- RNA is involved in **protein** synthesis.

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- Missing of enzymes homogentistic acid oxidase caused....**Alkaptonuria**
- Following has a major role in down syndrome..?**Maternal age**
- Enormous genetic information is present in...**DNA**
- Mandal concluded SS trait **dominant** to the ss trait (recessive).
- RFLP stands for **Restriction fragment length polymorphism.**
- **Electronegativity:** The tendency of an atom to attract electrons towards itself.
- Nucleoside contains **Pentose sugar + Nitrogenous base.**
- Nucleotide contains **Pentose sugar + Nitrogenous base + Phosphate Group**
- The structure of an amino acid is different due to its **R group.**

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- Crllobiose has a Beta **1, 4 glycosidic linkage**.
- Smooth endoplasmic reticulum functions in lipid and **steroid synthesis, detoxification, and calcium ion storage**.
- Cells are small to maintain a larger surface area **to volume ratio**.
- Ribosomes are produced in the **nucleolus** of the cell.
- DNA and proteins form a structure called **chromosomes**.
- Meiosis produces **four cells**.
- The phase between two mitotic cycles is called **interphase**.
- Mendel used **Peisum setiwum (Garden pea)** in his experiments on inheritance.
- Cytoskeleton consists of **microfilaments, intermediate filaments, and microtubules**.

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- Ammonia is formed at temperatures between **450 to 500** degrees Celsius.
- Okazaki fragments are joined by the **enzyme ligase**.
- In gel electrophoresis, the sample moves toward the **cathode (negative electrode)**.
- At equilibrium, delta G (Gibbs free energy change) **is 0**.
- The probability range is from **0 to 1**.
- Energy in motion is **kinetic energy**.
- The enzymatic site where the substrate attaches is called the **active site**.
- The shape of a protein is **three-dimensional**.
- The non-protein part of a protein is called the **prosthetic group**.
- Aquaporin's are protein channels that allow the passage of water through cell membranes by **osmosis**.
- In prokaryotes, the genetic material lies in the **nucleoid region**.

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- DNA is replicated during **the S phase** of the cell cycle.
- Different forms of genes are called **alleles**.
- DNA replication follows the **semi-conservative model**.
- SRP stands for **Signal recognition particles**.
- Ammonia is formed at temperatures between **450 to 500** degrees Celsius.
- Okazaki fragments are joined by the **enzyme ligase**.
- At equilibrium, delta G (Gibbs free energy change) **is 0**.
- The probability range is from **0 to 1**.
- The nuclear membrane is continuous with the **endoplasmic reticulum**.
- The **observable features** in the subject of genetics are called **traits, characters, and phenotypes**. All of these terms are used to describe different aspects of genetic characteristics.
- Cell membrane expels particles **by vesicle formation** through a process called **exocytosis**.

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- Eukaryotic ribosomes' large subunits contain **three different types of RNA molecules** (rRNAs): 28S, 5.8S, and 5S rRNA.
- The difference between alpha 1-4 glycosidic and beta 1-4 glycosidic bonds is based on the **size and shape** of the molecules involved.
- Maternal age has a major role in **Down syndrome**. Advanced maternal age, especially over 35 years old, increases the risk of having a child with Down syndrome.
- Polypeptide chains can form **both alpha helices and beta sheets**. They have the capability to adopt different secondary structures depending on the sequence of amino acids.
- Scale for Genetic map: **centimorgan (cM)**
- Amino acids divided into **2 groups**: polar and nonpolar (or hydrophobic and hydrophilic)

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- Mendel used which trait character: **pea plant traits**, such as seed color, plant height, flower color, etc.
- Unit of life: **cell**
- Phenylalanine hydroxylase changes into: **tyrosine**
- Proteins and nucleic acids made group is: **amino acids**
- SER assembled: **lipids**
- Messenger hypothesis makes protein by using: **ribosomes**
- Nucleotide made up of: **nitrogenous base, sugar, and phosphate**
- Process involved in prometaphase: **spindle fiber formation**, attachment of spindle fibers to kinetochores, and movement of chromosomes towards the center of the cell
- Kinesin and lysine are: **proteins**
- Agarose gel is obtained from: **seaweed or algae**

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- Step not involved in PCR: **gel electrophoresis (used for analyzing PCR products)**
- 2 nonpolar bonds interact: through **van der Waals forces** or hydrophobic interactions
- Which of the following are examples of monosaccharides?

Answer: a) **Ribose**

- Which molecule cannot carry a double bond?

Answer: b) **Saturated fat**

- The power generation organelle of the cell is:

Answer: c) **Mitochondria**

- Okazaki fragments are joined by:

Answer: a) **Ligase**

- Approximately how much water is present in a cell?

Answer: d) **70%**

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➤ Bacteriophage destroys which of the following?

Answer: c) **Bacteria**

➤ HIV is:

Answer: b) **RNA**

➤ Energy in motion is known as:

Answer: b) **Kinetic energy**

➤ Ribosomes are attached to:

Answer: c) **Endoplasmic reticulum**

➤ Which step is not involved in PCR (Polymerase Chain Reaction)?

Answer: d) **Initiation**

➤ In meiosis, a parent cell produces how many daughter cells?

Answer: d) **4**

➤ Polar amino acids:

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Zoology Legends

Answer: d) **Are hydrophilic**

➤ Separation on SDS-PAGE is based on:

Answer: a) **Charge and size**

➤ How many molecules does a cell membrane have?

Answer: d) **It varies depending on the cell type**

➤ The intake of material into a cell is called:

Answer: b) **Endocytosis**

➤ The space between the double membrane of the endoplasmic reticulum is called:

Answer: b) **Lumen**

➤ Okazaki fragments are joined by:

Answer: a) **Ligase**

➤ The nucleus was discovered by:

Answer: c) **Robert Brown**

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Zoology Legends

- Bacteriophages destroy:

Answer: c) **Bacteria**

- Prosthetic group for **Flavin is FMN or FAD**.
- A person with an extra Y chromosome has Klinefelter syndrome.
Answer: **Normal**
- Pulling of sister chromatids occurs in **Meiosis II**.
- Ribose is a **monosaccharide**.
- **Actin** stabilizes cell shape.
- The formation of NH_3 from H_2 and N_2 is carried out at a temperature range **of 450 to 500 °C**.
- The site on the ribosome where the charged **tRNA** lands with the help of the transfer factor is **the A site**.
- The region where a substrate binds to a molecule is called the **active site**.
- Examples of nucleic acids are **DNA and RNA**.
- VNTR stands for **Variable Number Tandem Repeats**.

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- The sugar in DNA is **deoxyribose**.
- Cells are small to maintain a large surface **area-to-volume ratio**.
- Hershey and Chase proposed that **DNA** is the hereditary material in **1944**.
- Okazaki fragments are joined by DNA **ligase**.
- The macromolecules of a cell are **proteins, nucleic acids, carbohydrates, and lipids**.
- The polymerase used in PCR is obtained from **Thermus aquaticus**.
- DNA carries a **negative charge**.
- **Taqman** is a probe present in **Real-Time PCR**.
- The trait "long life span" does not occur in Mendel's experiments with **Pisum sativum**.
- Robert Brown discovered the **nucleus**.
- Beta pleated sheet is found in the **secondary protein structure**.

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Zoology Legends

- DNA replication follows the **semi-conservative model**.
- The formation of NH_3 is carried out at a temperature of **450-500 °C**.
- The phase **G0** is not a phase of **interphase**.
- X chromosome is a **sex chromosome**.
- Observable features in the subject of genetics are collectively referred to as **phenotypes**.
- Inhibitors that bind to the active site are called **competitive inhibitors**.
- The inner membrane of mitochondria is called the **cristae**.
- Large subunits of ribosomes have **three binding sites for mRNA**.
- Which is prosthetic group.... **Flavin**
- Person with extra y chromosome is.. **Normal**
- Who discovered Nucleus--- **Robert Brown**

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Zoology Legends

- Beta pleated sheet is found in --- **Secondary protein structure**
- DNA replication--- **Semi- conservative model**
- Formation of NH_3 is carried out at ----- temperature.
450-500 °C
- Which is not phase of interphase ----- **phase G 0**
- Which one is a sex chromosome—**X**
- The observable feature in the subject of genetics called... **All correct**
- Inhibitors that binds to active site ... **Competitive inhibitors**
- Membrane in mitochondria---- **Cristae**
- Large subunits of ribosome has mRNA--- **3**
- Temperature during NH_3 formation.....**450-500C**
- What is important**maternal age**
- Okazaki fragments are joined by..... **DNA Ligase**

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- Translation takes place in the **cytoplasm and the nucleus.**
- Cell remove material by vesicle formation**exocytosis**
- **Arginine (Arg):** The UUU code for Arginine is **CGU**.
- **Proline (Pro):** The UUU code for Proline is **CCU**.
- **Alanine (Ala):** The UUU code for Alanine is **GCU**.
- DNA replicate during.....**s phase**
- Bacteriophage destroys..... **Bacteria**
- Power house of cell.....**mitochondria**
- Cell are small.....to maintain large surface **area to volume ratio**
- Which is not cofactor.....**NAD**
- Substrate attach to**active site**
- The law of independent assortment states that **alleles segregate independently** of each other during **gamete formation,**

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Zoology Legends

- DNA me WO Jo PCR me polymerase ki length thi is trha tha kuch options me 30-32, 10-14, 12-24 option thy
- Starch and glycogen are found in **plants and animals**, while glucose is found in **plants or herbs**.
- Monosaccharides: **Glucose, fructose, and ribose** are examples of monosaccharides.
- A nucleoside is a simple sugar molecule bonded to a **nitrogenous base**.
- The **activation energy** of an enzyme is the energy barrier that must be overcome for a specific chemical reaction to occur.
- **Sex-related** disorders are conditions that affect the development or functioning of the **reproductive system**.
- Down syndrome is primarily caused by the presence of an extra copy of **chromosome 21**.

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- Different properties are given to each of the amino acids by their unique side chains or **R-groups**.
- Amino acids can be free (unbound) or attached to **tRNA** molecules **during protein synthesis**.
- The function of the Golgi apparatus was
- ΔV in a reaction refers to the **change in volume**.
- The movement of electrons itself is represented by a **negative sign**.
- **Endocytosis** is the process by which cells internalize substances from the external environment by engulfing them.
- The observable fracture in the subject of genetics can refer to **genetic mutations or genetic disorders**.
- Enzyme nature: **Protein**
- Nucleus was discovered by: **Robert Brown**
- Double helix structure of DNA was discovered in: **1953**

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- Which molecule is used as a template for protein synthesis: **mRNA**
- Endonuclease was discovered by: **Richard J. Roberts**
- The Golgi apparatus functions to modify, process, sort, and package proteins and lipids synthesized in the endoplasmic reticulum before transporting them to their final destinations within the cell.
- Transfer of electrons from one atom to another results in which type of bond?
- Answer: **b) Ionic bond**

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Zoology Legends

Repeated Subjective

1. What is the threshold effect on identical twins? (3 marks)

A disorder manifests itself only when a certain number of effector genes, as well as conditioning environmental influences are involved this phenomenon is termed as threshold effect

Recurrence rate 2%-7% in for first degree relatives and 20%-40% for identical twins.

Twins are two offspring produced by the same pregnancy. Twins can be either monozygotic ("identical"), meaning that they develop from one zygote, which splits and forms two embryos, or dizygotic ("fraternal"), meaning that they develop from two different eggs.

2. What is the difference between purines and pyrimidines? (3 marks)

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Purines	Pyrimidines
The purines in DNA are adenine and guanine, the same as in RNA.	The pyrimidines in DNA are cytosine and thymine in RNA, they are cytosine and uracil.
Purines are larger than pyrimidines because they have a two-ring structure.	Pyrimidines only have a single ring.

3. What are the requirements for PCR? (3 marks)

- Target DNA
- Two oligonucleotide primers
- DNA polymerase,
- dNTPs
- PCR machine

(Agr 5 Marks Ka Pouchay To AP In KO Explain Kr Dain)

4. What are the requirements (Componentes) for PCR? (5 marks)

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DNA template: The DNA sample that contains the target sequence to be amplified.

Primers: Short DNA sequences that flank the target region and initiate DNA replication.

DNA polymerase: Enzyme responsible for synthesizing new DNA strands during amplification.

Nucleotides (dNTPs): Building blocks of DNA that are used to extend the primers and generate new DNA strands.

Buffer solution: Provides the optimal pH and ionic conditions for the PCR reaction.

Magnesium ions (Mg^{2+}): Required cofactor for DNA polymerase activity.

Thermal cycler: Instrument that controls the temperature cycles for denaturation, annealing, and extension.

PCR tubes or plates: Containers that hold the reaction mixture during the PCR process.

Thermal cycling conditions: Specific temperature and time settings for denaturation, annealing, and extension steps.

PCR additives (optional): Additional components such as stabilizers, enhancers, or inhibitors that may be added to improve the PCR reaction.

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5. Explain the difference between co-factor, co-enzyme, and prosthetic group with examples. (5 marks)

Cofactors: Inorganic ions (Cu, Zn & Fe) bound to some enzymes and essential to their function.

Coenzymes: Carbon containing molecules bind enzymes temporarily chemically changed during reaction (substrate).

Prosthetic groups: nonprotein component permanently bound to enzymes (heme group), without its prosthetic group apoprotein vs. holoprotein.

6. What is the difference between DNA and RNA? (5 marks)

Difference	RNA	DNA
Structure:	Usually single-stranded and can fold upon itself.	Double-stranded helical structure with two complementary strands.
Sugar:	Contains ribose sugar.	Contains deoxyribose sugar.

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Function:	Plays a crucial role in protein synthesis, acts as an intermediary molecule.	Carries genetic information, serves as long-term storage, provides instructions for RNA and protein synthesis.
Stability:	Generally less stable than DNA.	Relatively stable and withstands environmental conditions.
Presence:	Found in the nucleus, cytoplasm, and various organelles.	Found in the nucleus and mitochondria.

7. List the functions of DNA. (5 marks)

The function of DNA is to store all of the genetic information that an organism needs to develop, function, and reproduce.

DNA is involved in:

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- Genetic information storage.
- Transmission of genetic information.
- Protein synthesis.
- Regulation of gene expression.
- DNA repair and maintenance.
- Replication process
- Mutations
- Transcription
- Cellular Metabolism
- DNA Fingerprinting
- Gene Therapy

8. Define hypotonic, hypertonic, and isotonic environments.

Hypotonic: An environment where the concentration of solutes outside a cell is lower than the concentration inside the cell. In a hypotonic environment, water moves into the cell, causing it to swell or potentially burst.

Hypertonic: An environment where the concentration of solutes outside a cell is higher than the concentration inside the cell. In a hypertonic environment, water moves out of the cell, causing it to shrink or shrivel.

Isotonic: An environment where the concentration of solutes outside a cell is equal to the concentration inside the cell. In an

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isotonic environment, water moves in and out of the cell in equal amounts, resulting in no net change in cell volume.

9. What are nucleotides and nucleosides?

Nucleotides:	Nucleosides:
Nucleotides are the building blocks of nucleic acids, such as DNA and RNA. Each nucleotide consists of three components: a nitrogenous base (adenine, guanine, cytosine, or thymine/uracil), a sugar molecule (ribose in RNA or deoxyribose in DNA), and a phosphate group. Nucleotides are linked together through phosphodiester bonds to form the backbone of DNA and RNA strands.	Nucleosides are similar to nucleotides but lack the phosphate group. They consist of a nitrogenous base and a sugar molecule (ribose or deoxyribose). Nucleosides can be phosphorylated by adding a phosphate group to form nucleotides. Nucleosides serve as essential components in various biological processes, including energy transfer (such as adenosine triphosphate, ATP) and as building blocks for nucleic acid synthesis

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10. Name the components of cellular membranes.

Components of cellular membranes:

Phospholipids: The primary structural component of cellular membranes. They consist of a hydrophilic (water-loving) head and hydrophobic (water-repelling) tails, forming a lipid bilayer that provides the barrier of the membrane.

Proteins: Integral and peripheral proteins are embedded within or attached to the lipid bilayer. These proteins serve various functions, including transport of molecules, cell signaling, enzymatic activity, and structural support.

Cholesterol: Present in animal cell membranes, cholesterol helps regulate the fluidity and stability of the membrane. It inserts itself among phospholipids, influencing membrane fluidity and preventing excessive rigidity or fluidity.

Carbohydrates: Found on the outer surface of the cell membrane, carbohydrates attach to lipids (glycolipids) or proteins (glycoproteins) to form glycolipids and glycoproteins. They participate in cell recognition, cell adhesion, and cell signaling processes.

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11. Explain the principle of RFLP.

The principle of RFLP (Restriction Fragment Length Polymorphism) is based on the fact that individuals have variations in the DNA sequences recognized by restriction enzymes. RFLP analysis involves the following steps:

DNA Extraction: DNA is extracted from the sample (e.g., blood, tissue) of interest.

Restriction Enzyme Digestion: The DNA sample is treated with specific restriction enzymes that recognize and cut DNA at specific nucleotide sequences, resulting in the formation of DNA fragments.

Gel Electrophoresis: The DNA fragments are separated based on their size using gel electrophoresis, a technique where an electric current is applied to a gel matrix, causing the DNA fragments to migrate through the gel based on their size.

DNA Fragment Detection: After electrophoresis, the DNA fragments are visualized using staining methods or labeled probes. The resulting pattern of DNA fragments represents the unique genetic profile or fingerprint of an individual.

Analysis and Interpretation: The DNA fragment pattern is compared among different individuals to identify variations in the DNA sequence. Variations in the size or presence/absence of

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DNA fragments indicate genetic polymorphisms, which can be used for genetic analysis, such as in forensic investigations or paternity testing.

12. What is the difference between aneuploidy and polyploidy?

Aneuploidy	Polyploidy
Aneuploidy refers to a condition where there is an abnormal number of chromosomes in a cell. It occurs when there is a loss or gain of individual chromosomes, resulting in an imbalance. For example, trisomy 21 (Down syndrome) is a form of aneuploidy where there is an extra copy of chromosome 21.	Polyploidy, on the other hand, refers to a condition where there are multiple complete sets of chromosomes in a cell. It occurs when there is an entire extra set of chromosomes. Polyploidy is often denoted as triploidy (three sets), tetraploidy (four sets), or higher multiples, such as hexaploidy (six sets) or octaploidy (eight sets).

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13. Describe the function of cellular membranes.

Cellular membranes function as:

- Selective barriers, controlling the movement of substances in and out of cells.
- Facilitators of cell signaling, allowing communication between cells and their environment.
- Maintainers of cell integrity, providing physical protection and separating internal components from the external environment.
- Regulators of transport, facilitating the movement of molecules and ions across the membrane.

14. What are the features of mitochondria?

Features of mitochondria include:

- Double membrane structure (outer and inner membrane)
- Powerhouses of the cell, producing energy (ATP)
- Presence of mitochondrial DNA (mtDNA)
- Endosymbiotic origin
- Involved in calcium homeostasis
- Mediate cell growth and death
- Play a role in programmed cell death (apoptosis)

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15. Define plasmid, mutation, and genetic engineering.

Plasmid: A small, circular DNA molecule in bacteria and other organisms that replicates independently. It can carry additional genes, such as antibiotic resistance genes, which provide advantages to the host organism.

Mutation: A change in the DNA sequence of an organism's genome.

Examples include point mutations (single base changes), insertions, deletions, or rearrangements. Mutations can lead to genetic variations and phenotypic changes, such as altered protein function or disease susceptibility.

Genetic Engineering: The manipulation of an organism's genetic material using techniques like recombinant DNA technology.

Examples include the insertion of a gene from one organism into another to confer a desired trait, such as insect resistance in crops, or the modification of genes to produce therapeutic proteins like insulin.

16. Define heterozygous, homozygous, phenotype, and genotype.

Heterozygous: Heterozygous refers to having two different alleles (alternate forms of a gene) at a specific genetic locus.

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For example, if an individual inherits one allele for brown eyes and one allele for blue eyes, they would be heterozygous for the eye color gene.

Homozygous: Homozygous refers to having two identical alleles at a specific genetic locus.

For example, if an individual inherits two alleles for brown eyes, they would be homozygous for the eye color gene.

Phenotype: Phenotype refers to the observable traits or characteristics of an organism, resulting from the interaction between its genotype (genetic makeup) and the environment.

Examples of phenotypic traits include physical appearance, behavior, and biochemical properties.

Genotype: Genotype refers to the genetic makeup of an organism, specifically the combination of alleles present at a particular genetic locus. It represents the genetic information carried by an individual, which may or may not be expressed in the phenotype.

17. What are the types of cristae in cells?

The types of cristae in cells are:

Tubular cristae: These cristae have a tubular or finger-like shape, extending into the matrix of the mitochondria.

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They provide a large surface area for ATP synthesis and are commonly found in mitochondria involved in energy-intensive processes, such as muscle cells.

Lamellar cristae: These cristae have a flat, sheet-like structure that runs parallel to the inner mitochondrial membrane.

They are found in mitochondria involved in lipid metabolism and other biochemical processes.

18. Explain the process of semi-conservative replication, dispersive replication, and conservative replication.

☐ **Semiconservative replication**

Semiconservative replication describes the mechanism of DNA replication in all known cells.

DNA replication occurs on multiple origins of replication along the DNA template strand. As the DNA double helix is unwound by helicase, replication occurs separately on each template strand in antiparallel directions.

☐ **Conservative Replication**

In conservative replication, two DNA copies are produced from one original DNA, which serves as a template. Out of these two, one is entirely new DNA, and the other is made of old DNA strands.

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□ Dispersive replication

It is a replication process that also produces two DNA copies from the original DNA. Here, the two DNA products have distinct regions composed of new and original strands. This replication happens in a random fashion.

19. Differentiate between symporter, antiporter, and uniporter.

Importers: Unidirectional transport proteins that facilitate the movement of specific molecules across cellular membranes.

Symporters: Transport proteins that simultaneously move two different molecules in the same direction, using the energy generated by the electrochemical gradient of one molecule.

Antiporters: Transport proteins that move two different molecules in opposite directions, harnessing the energy from concentration gradients or membrane potential.

To differentiate:

- Importers transport a single molecule in one direction.
- Symporters transport two molecules in the same direction.
- Antiporters transport two molecules in opposite directions.

The direction and energy coupling mechanism distinguish these transporters in cellular transport processes.

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20. What is the purpose of using dye in gel electrophoresis?

The purpose of using dye in gel electrophoresis is to visualize the DNA or protein bands. Dyes, such as ethidium bromide or fluorescent dyes, bind to the DNA or protein molecules and emit fluorescence when exposed to UV light or when excited by a laser.

This allows researchers to see the bands of separated DNA or protein fragments in the gel, aiding in the analysis and identification of specific molecules.

21. What are the steps of meiosis II?

The steps of meiosis II are:

Prophase II: The nuclear envelope breaks down, and chromosomes condense.

Metaphase II: Chromosomes align at the metaphase plate of the cell.

Anaphase II: Sister Chromatids separate and move towards opposite poles.

Telophase II: Nuclear envelopes reform, and four haploid daughter cells are produced.

22. What is the role of proteins?

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The role of proteins includes:

- Structural support for cells and tissues.
- Enzymes that catalyze chemical reactions.
- Transporters for molecules across cellular membranes.
- Regulators of gene expression and cell signaling.
- Defense mechanisms through antibodies in the immune system.
- Energy production and storage.
- Cell adhesion and communication.
- Maintenance of pH balance and osmotic regulation.
- Muscle contraction and movement.
- Hormone production and regulation.
- Maintenance of cell shape and organization.

23. Explain the cell cycle briefly.

The cell cycle is the series of events that take place in a cell leading to its division and the production of two daughter cells.

It consists of interphase, which includes the G1 phase (cell growth), S phase (DNA replication), and G2 phase (preparation for division), followed by the mitotic phase, which includes mitosis (nuclear division) and cytokinesis (cytoplasmic division).

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The cell cycle ensures growth, repair, and reproduction of cells and is tightly regulated to maintain proper cell function and prevent uncontrolled cell division.

24. What is phenylketonuria (PKU)?

Phenylketonuria (PKU) is a genetic disorder characterized by the inability to properly metabolize an amino acid called phenylalanine.

It is caused by a deficiency of the enzyme phenylalanine hydroxylase, which leads to the accumulation of phenylalanine in the body. This can result in intellectual disability, developmental delays, and other neurological problems if not treated early through a special diet low in phenylalanine.

PKU is typically identified through newborn screening tests.

PAH: is the enzyme which causes PKU

25. List the organelles of a cell.

The organelles of a cell include:

- Nucleus
- Mitochondria
- Endoplasmic reticulum
- Golgi apparatus
- Lysosomes

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- Peroxisomes
- Vacuoles
- Chloroplasts (found in plant cells)
- Ribosomes
- Cytoskeleton

26. What is the difference between endocytosis and exocytosis?

Endocytosis	Exocytosis
Endocytosis is the process by which cells take in substances from the external environment by engulfing them with their cell membrane. It forms a vesicle inside the cell, allowing the internalization of materials.	Exocytosis, on the other hand, is the process by which cells release substances to the external environment. It involves the fusion of vesicles containing materials with the cell membrane, leading to their secretion outside the cell.

27. Explain the applications of PCR.

The polymerase chain reaction (PCR) is a widely used molecular biology technique with various applications, including:

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DNA amplification: PCR allows for the rapid and specific amplification of DNA sequences, enabling the generation of large quantities of DNA from a small starting amount.

It is used in various fields, such as genetic research, diagnostics, forensics, and paternity testing.

Genetic testing: PCR is utilized in genetic testing to detect genetic mutations, diagnose genetic diseases, and screen for genetic disorders.

It enables the identification of specific DNA sequences associated with genetic conditions.

Disease diagnosis: PCR is used in medical diagnostics to detect and identify infectious agents such as bacteria, viruses, and parasites.

It aids in the diagnosis of diseases by detecting the presence of specific DNA or RNA sequences associated with the pathogen.

Forensic analysis: PCR is employed in forensic science to analyze DNA evidence collected from crime scenes.

It enables the identification and comparison of DNA profiles, helping in the identification of suspects or establishing relationships between individuals.

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DNA sequencing: PCR plays a crucial role in DNA sequencing methods, such as Sanger sequencing and next-generation sequencing.

It is used to amplify specific DNA regions for sequencing, allowing the determination of the order of nucleotides in a DNA molecule.

Cloning and genetic engineering: PCR is employed in cloning and genetic engineering techniques to amplify specific DNA fragments for subsequent manipulation and insertion into other organisms.

It aids in the construction of recombinant DNA molecules and the production of genetically modified organisms.

28. What is the role of lipids?

The role of lipids includes:

- **Energy storage:** Lipids serve as a long-term energy storage molecule in the body. They store more than twice the amount of energy per gram compared to carbohydrates.
- **Structural component:** Lipids are an essential component of cell membranes, providing structural integrity and acting as a barrier between the cell and its environment.

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- **Insulation and protection:** Lipids, such as adipose tissue, help insulate the body and provide protection to organs by cushioning them against physical impact.
- **Hormone production:** Lipids play a crucial role in the production of hormones, including steroid hormones like estrogen and testosterone, which regulate various physiological processes.
- **Absorption of fat-soluble vitamins:** Lipids aid in the absorption and transportation of fat-soluble vitamins (A, D, E, and K) in the body.
- **Thermal regulation:** Lipids, particularly in the form of adipose tissue, help regulate body temperature by acting as an insulating layer.

29. Explain the transcription process.

The transcription process can be summarized as follows:

Initiation: RNA polymerase binds to the promoter region on DNA, marking the starting point for transcription. The DNA double helix unwinds, forming a transcription bubble.

Elongation: RNA polymerase synthesizes a complementary RNA strand using one of the DNA strands as a template. It adds

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complementary RNA nucleotides (adenine, cytosine, guanine, and uracil) according to the base pairing rules.

Termination: Transcription continues until a termination signal is reached on the DNA template. The RNA polymerase and newly synthesized RNA molecule are released from the DNA.

Post-transcriptional processing: In eukaryotes, the initial RNA transcript, called pre-mRNA, undergoes several modifications, including the removal of introns (non-coding regions) and the addition of a 5' cap and a poly-A tail.

These modifications prepare the mRNA for translation.

30. Describe lipids. (3 marks)

Lipids are a diverse group of biomolecules that are hydrophobic (insoluble in water). They include fats, oils, phospholipids, and steroids.

Lipids serve as a long-term energy storage, provide insulation and protection to organs, act as structural components of cell membranes, and serve as signaling molecules.

They are composed of fatty acids and glycerol or other nonpolar molecules.

31. What are the main dilemmas of cell replication?

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Main dilemmas of cell replication:

- **DNA fidelity:** Ensuring accurate replication without errors or mutations.
- **Replication timing:** Coordinating replication with other cellular activities.
- **Replication completion:** Ensuring full genome replication before cell division.
- **Telomere maintenance:** Preserving the integrity of chromosome ends.
- **Replication stress:** Managing challenges like DNA damage or replication fork stalling.
- **Replicative aging:** Coping with the gradual loss of replicative capacity over time.

32. Explain the principle of gel electrophoresis.

Gel electrophoresis separates DNA, RNA, or proteins based on size and charge using an electric field and gel matrix.

- **Sample loaded:** Molecules of interest are prepared and loaded onto the gel.
- **Gel matrix:** Gel (agarose or polyacrylamide) slows migration based on size.

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- Electric field: Applied across the gel, molecules migrate towards the positive end.
- Separation: Smaller molecules move faster and travel further, while larger molecules move slower.
- Visualization: Separated molecules are visualized using dyes, stains, or probes.
- Analysis: Molecules can be further analyzed, quantified, or purified based on their positions in the gel.

33. What are the steps of mitosis?

The steps of mitosis are:

- **Prophase:** Chromosomes condense, and the nuclear membrane breaks down.
- **Metaphase:** Chromosomes line up along the equator of the cell.
- **Anaphase:** Sister Chromatids separate and move to opposite poles.
- **Telophase:** Chromosomes reach the poles, and the nuclear membrane reforms.
- **Cytokinesis:** The cell divides into two daughter cells.

34. What is the role of protein in living organisms?

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The role of proteins in living organisms includes:

- **Enzymatic activity:** Proteins serve as enzymes, catalyzing biochemical reactions in cells.
- **Structural support:** Proteins form the structural components of cells, tissues, and organs.
- **Transport and storage:** Proteins transport molecules across cell membranes and store essential substances.
- **Cell signaling:** Proteins play a crucial role in transmitting signals within and between cells.
- **Immune defense:** Proteins participate in immune responses, recognizing and eliminating foreign substances.
- **Regulation:** Proteins regulate gene expression, cell cycle progression, and metabolic pathways.

35. Name three RNA viruses.

Three examples of RNA viruses are:

- **Influenza virus:** The virus responsible for causing influenza (flu) in humans and other animals.
- **Hepatitis C virus:** A virus that infects the liver and can lead to chronic liver disease.

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- **Measles virus:** A highly contagious virus that causes measles, a respiratory infection characterized by fever, rash, and other symptoms.

36. Define euchromatin and heterochromatin.

Euchromatin: Euchromatin refers to the loosely packed and transcriptionally active form of chromatin. It contains genes that are actively transcribed and is associated with a more open chromatin structure.

Heterochromatin: Heterochromatin refers to the densely packed and transcriptionally inactive form of chromatin. It contains genes that are typically silenced and is associated with a more condensed chromatin structure.

37. Name the three types of codons.

The three types of codons are:

Start codon: The start codon, specifically the codon AUG (adenine-uracil-guanine), signals the beginning of protein synthesis. It also codes for the amino acid methionine.

Stop codons: There are three stop codons in the genetic code: UAA (uracil-adenine-adenine), UAG (uracil-adenine-guanine), and UGA (uracil-guanine-adenine). These codons signal the termination of protein synthesis and do not code for any amino acids.

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Sense codons: Sense codons encode specific amino acids and are responsible for translating the genetic information into protein sequences. There are 61 sense codons in the genetic code, each coding for one of the 20 standard amino acids.

38. Types of cristae Golgi apparatus

- Cristae Membrane
- Cristae Junctions

39. What is the larger subunit of ribosomes?

The larger subunit of ribosomes is called the 50S subunit.

40. Explain feedback inhibition with the Krebs cycle.

Feedback inhibition is a regulatory mechanism in which the end product of a metabolic pathway inhibits an earlier step in the pathway.

In the Krebs cycle (also known as the citric acid cycle or TCA cycle), feedback inhibition occurs when the final product, ATP, binds to and inhibits the enzyme isocitrate dehydrogenase.

This helps to regulate the cycle and prevent excessive ATP production when energy demands are met.

41. What are the models of DNA replication?

The two models of DNA replication are the semi-conservative model and the conservative model.

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Semi-conservative model: According to this model, during DNA replication, the two strands of the parental DNA molecule separate, and each strand serves as a template for the synthesis of a new complementary strand.

The resulting DNA molecules consist of one original parental strand and one newly synthesized strand.

Conservative model: This model suggests that during DNA replication, the parental DNA molecule remains intact, and a completely new molecule is synthesized.

In this process, both strands of the parental DNA serve as templates for the formation of two entirely new strands, resulting in two DNA molecules, one consisting of the original parental DNA and the other entirely newly synthesized.

42. What molecules make up amino acids?

Amino acids are composed of three main molecules: an amino group ($-NH_2$), a carboxyl group ($-COOH$), and a side chain (R group).

43. Explain the physical barrier of enzyme regulation.

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The physical barrier of enzyme regulation refers to the presence of physical structures or compartments that restrict the access of enzymes to their substrates or modulators.

These barriers can include cellular membranes or compartments that separate enzymes from their targets, limiting their interaction and regulating their activity.

44. Explain the messenger and adapter hypothesis.

The messenger and adapter hypothesis proposes the mechanism by which genetic information is translated into proteins. Here are the main points:

Messenger Hypothesis: According to this hypothesis, the genetic information stored in DNA is transcribed into messenger RNA (mRNA) molecules.

mRNA carries the genetic code from the DNA in the nucleus to the ribosomes in the cytoplasm.

Adapter Hypothesis: The adapter hypothesis suggests that specific molecules called transfer RNA (tRNA) act as adapters or intermediaries during protein synthesis.

Each tRNA molecule binds to a specific amino acid and recognizes a specific sequence of mRNA codons through its anticodon region.

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Codon-Anticodon Interaction: The mRNA molecule contains a sequence of codons, which are three-nucleotide sequences that specify a particular amino acid.

The tRNA molecules with complementary anticodons bind to the corresponding codons on the mRNA during translation.

Protein Synthesis: As the ribosome moves along the mRNA, amino acids carried by tRNA molecules are added to the growing polypeptide chain according to the codon sequence.

This process continues until a stop codon is encountered, signaling the termination of protein synthesis.

45. Differentiate between the each Step of PCR:

Denaturation: This is the First regular cycling event and consists of heating the reaction to 93°C - 98°C for 20- 45 seconds.

It causes melting of DNA template yielding single strands of DNA.

Annealing: In this reaction temperature is lowered to 40-65°C for 20-40 seconds.

Annealing of the primers to the single stranded DNA template.

Extension: At this step the DNA polymerase synthesizes a new DNA strand complementary to the DNA template strand by

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adding dNTP's. The temperature at this step depends on the type of DNA polymerase used.

Taq polymerase has its optimum activity temperature at 70-80°C.

Commonly 72°C is used.

Muhammad Zaman

03044756496

Remember me in you Precious Prayer

Jazak Allah

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